



Document Management System



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Title: Vancomycin Therapeutic Dosing and Monitoring in Children

Introduction:

- Vancomycin is a glycopeptide antibiotic which acts by inhibition of bacterial cell wall synthesis. It is bactericidal against staphylococci and most streptococci, and bacteriostatic against enterococcus.
- Half life of vancomycin:- range from 5 to 10 hours in newborns
 - 4 hours in older infants
 - 2 to 3 hours in older children
 - anuric children up to 7 days
- 90% of vancomycin excreted by kidneys
- Vancomycin is approximately 55% bound to serum protein

Dosing:

Starting doses:

- Neonates 34 to 44 weeks post conceptional age: 25mg/kg/dose every 12 hours

Dosing Regimen	Age Criteria
70 mg/kg/day Q6hrs	3 months to < 2 years old
70 mg/kg/day Q6hrs	≥2 years old, Scr< 0.45** ((0.45 mg/dl = 40 micromoles/ L))

60 mg/kg/day Q6hrs

((0.45 mg/dl = 40 micromoles/ L))

** Could also consider this dosing if 30% or more MRSA isolates have MIC ≥ 1.5

Dosing considerations:

- Obese patients: Use actual body weight for calculation of initial Vancomycin dose.
- Meningitis: Give every 6 hours for meningitis with Gram positive cocci.

- **Schwartz equation:** $\text{CrCl (ml/min/1.73m}^2\text{)} = [\text{length (cm)} \times k] / \text{Scr}$
- (Patient population: infants over 1 week old through adolescence (18 years old))
- $k = 0.45$ for infants 1 to 52 weeks old
- $k = 0.55$ for children 1 to 13 years old
- $k = 0.55$ for adolescent females 13-18 years old
- $k = 0.7$ for adolescent males 13-18 years old
- ((0.45 mg/dl = 40 micromoles/ L))

Dosing and Monitoring if e GFR is 90mL/min/1.73m² or greater:

Indication	Dose
In critically ill patients including <i>staph.</i> Sepsis/MRSA bacteremia/ suspected central line infections	Loading dose 20mg/kg (max 1.5g) followed by maintenance dose of 15mg/kg (max 750mg) 6-hourly. The maintenance dosing should be commenced 6 hours after the loading

Dosing and Monitoring if e GFR is less than 90mL/min/1.73m²:

Level of renal function	Dose	Frequency	Levels
GFR 50-90 mL/min/1.73m ²	15mg/kg (max 1g)	8-hourly	Pre-dose level after 24 hours of treatment and withhold further doses until result known. ☐ _ Do not increase dose if level is low, but re-check after a further 24 hours of treatment. If levels still low at 48 hours, then adjust the dose as table below. ☐ _ If level high, follow guidance below.

			hours.
GFR 20 –50 mL/min/1.73m ²	7.5mg/kg	24 hourly	Before second dose is due: withhold dose until level known.
GFR < 20 mL/min/1.73m ² and/or patients on dialysis	7.5mg/kg	ONE DOSE ONLY given after dialysis in those on	Check level at 24 hours post dose or after next dialysis session. Do not give further dose until level known to have reduced to target

Target level:

- The target level is 10-15mg/L, or 15-20mg/L if complicated infection secondary to *S.aureus* (bacteremia with shock state, endocarditis, osteomyelitis, meningitis, and suspected PVL positive MRSA infection).
- If levels are in range, further levels should be checked every 3-4 days.
- If the levels are not in range, adjust as below (under Dose Adjustments)

Time to steady state: 1-2 days.

Administration:

- Vancomycin can be given by Peripheral or Central Intravenous Infusion.
- If treating infection of a multi-lumen central line consider dividing the dose between the lumens or giving doses via alternate lumens.
- Vancomycin must be infused over at least 1 to 2 hours : doses over 600mg should be given at a maximum rate of 10mg/minute.
- Monitor for adverse reactions throughout infusion, particularly for “Red-man” Syndrome: flushing of skin, severe hypotension, wheezing and dyspnoea.

Dose Adjustments:

- Vancomycin exhibits linear pharmacokinetics; an increase or decrease in dose should result in a proportionate increase or decrease in plasma concentrations (in stable renal function).

Sampling time:trough level (blood sample taken 30 minutes prior to dose) on day 2 of therapy (ideally just before the fourth dose to coincide with achievement of steady state).

Trough level sampling time:

Dose frequency	Timing of trough level

	12 hourly dosing	30 minutes before the 3rd or 4th dose
	8 hourly dosing	30 minutes before the 4th or 5th dose
	6 hourly dosing	30 minutes before the 5th or 6th dose

After dose change is indicated:

- Retest trough level 24 hours after dose change (trough level pre fourth dose).
- Trough level: Blood sample should be taken 30 minutes BEFORE dose is due.

When dose is within the target level:

- For **critically ill OR renal impaired**, repeat trough level in approximately one-day time.
- For non-critically ill patients (with stable renal function), recheck trough level until there are two consecutive level within the target level for that infection then level can be repeated every 3 to 4 days whenever there is a significant change in bodyweight, serum creatinine or if the dose has been adjusted.

If aiming for level of 10-15mg/mL (Standard)		
Level (mg/mL)	Action	When to do next level
<8	Increase the frequency to add an extra dose in each 24 period (e.g. change from 12-hourly to 8-hourly). If already on 6-hourly dosing, increase dose by 10-20% but do not increase frequency.	After 24 hours
8-10	Increase dose by 20%	After 24 hours
10-15	No change needed	In 3 - 4 days, assuming renal function remains stable.
15-18	Reduce dose by 10%	After 24 hours

18-22	dose in each 24-hour period e.g. reduce from 8-hourly to 12-hourly.	After 24 hours
>22	Check renal function and wait for trough to fall into range before giving any further doses. Once level is back in range, recommence dosing with an increased interval (e.g. change from 8-hourly to 12-hourly). Re-check level after 24 hours of treatment at new dose.	

If aiming for level of 15-20mg/mL (If deep-seated infection, or resistant strain of MRSA)		
Level (mg/mL)	Action	When to do next level
<10	Increase the frequency to add an extra dose in each 24 period (e.g. change from 12-hourly to 8-hourly). If already on 6-hourly dosing, increase dose by 20% but do not increase frequency.	After 24 hours
10-15	Increase dose by 10%	After 24 hours
15-20	No change needed	In 3 - 4 days, assuming renal function remains stable.
20-22	Reduce dose by 10%	After 24 hours
>22	Check renal function and wait for trough to fall into range before giving any further doses. Once level is back in range, recommence dosing with an increased interval (e.g. change from 8-hourly to 12-hourly). Re-check level after 24 hours of treatment at new dose.	

Toxicity:

A- Two most important adverse effects are nephrotoxicity and ototoxicity. Both relate to serum drug concentration and duration of therapy. The rate of uptake of vancomycin into the renal tissue is related to the time of exposure to higher levels rather than the absolute concentration.

B- Red man syndrome, can occur if infusion is too rapid. symptoms of which include flushing of skin, severe hypotension, wheeze and dyspnea.

If patient develop red man syndrome;

1- stop infusion

3-reassume the infusion at reduced rate

4-Infuse subsequent doses over 90 – 120 minutes and consider administration of an antihistamine before future doses

C- Other side effects include: rash and blood disorders such as neutropenia.

Reference:

1. Intravenous Vancomycin policy, Andrew Wignall, Nottingham University Hospital, Febraury 2014
2. Pediatrics Vancomycin Therapeutic Drug Monitoring, Queensland Government, 06/2014
3. Lee et al, Pediatric Infect Dis J, 2013
4. https://www.rch.org.au/clinicalguide/guideline_index/Vancomycin/, November, 2019

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